Lecture 6: Sections 1.5 and 1.6

Squeeze Theorem and Continuity

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Squeeze Theorem (Sandwich Theorem)

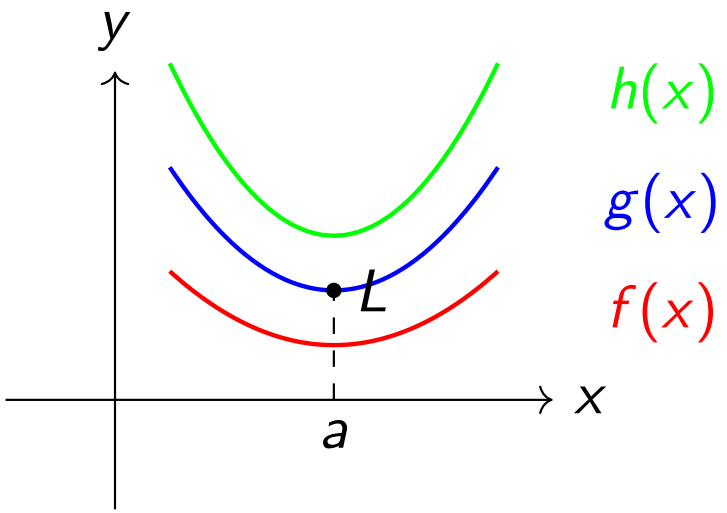
Theorem Statement

If $f(x) \leq g(x) \leq h(x)$ for all x near a (except possibly at a), and

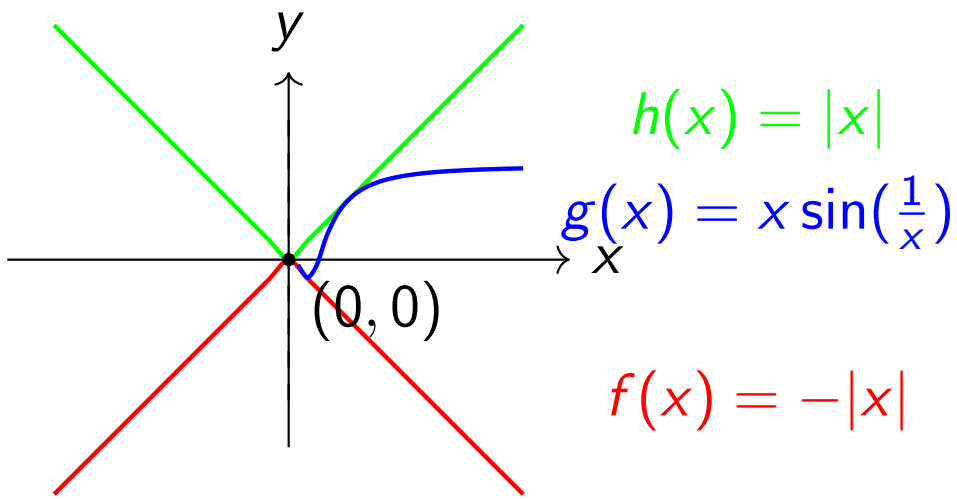
$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L$$



Visualizing the Squeeze Theorem



Since $-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$ for all $x \neq 0$ and

$$\lim_{x \to 0} -|x| = \lim_{x \to 0} |x| = 0$$

then by the Squeeze Theorem:

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$$

Squeeze Theorem Applications

Example

Use the Squeeze Theorem to evaluate:

1. $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$
2. $\lim_{x \to 0} x^2 e^{\sin(1/x)}$
3. $\lim_{x \to \infty} \frac{\sin x}{x}$

Step-by-Step: $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1 \quad \text{for all } x \neq 0$$

$$-x^2 \leq x^2 \cos\left(\frac{1}{x}\right) \leq x^2$$

$$\lim_{x\to 0} -x^2 = 0, \quad \lim_{x\to 0} x^2 = 0$$

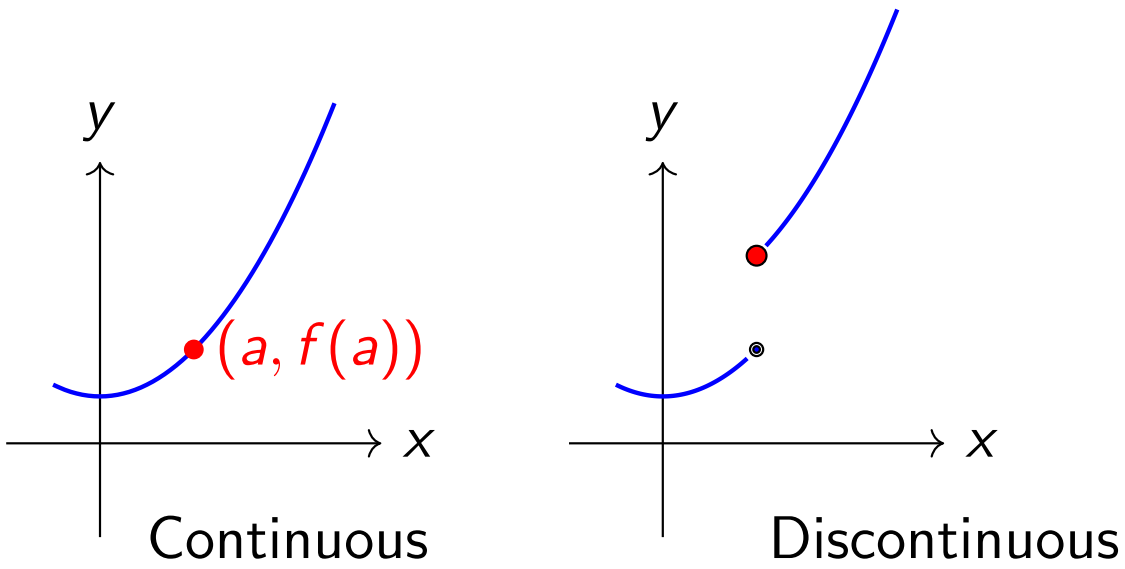
By Squeeze Theorem: $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right) = 0$

Definition of Continuity

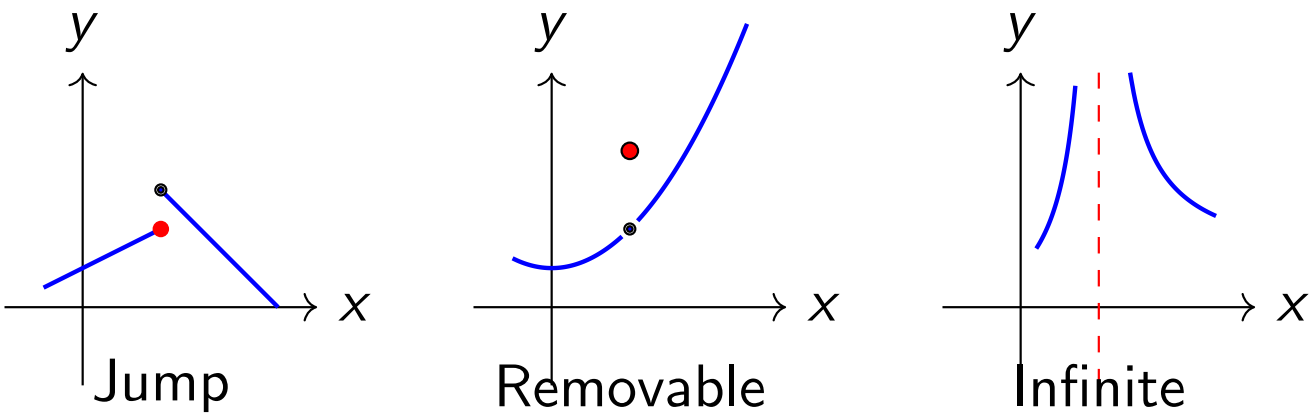
A function f is continuous at $x = a$ if:

1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exists
3. $\lim_{x \to a} f(x) = f(a)$

A function is continuous on an interval if it is continuous at every point in that interval.



Types of Discontinuity



Classification

- **Removable:** Limit exists but $f(a)$ is wrong/undefined
- **Jump:** Left and right limits exist but are different
- **Infinite:** Function approaches $\pm\infty$

Basic Properties of Continuous Functions

Algebraic Properties

If f and g are continuous at $x = a$, then:

- $f + g$ is continuous at a
- $f - g$ is continuous at a
- $f \cdot g$ is continuous at a
- $\dfrac{f}{g}$ is continuous at a (if $g(a) \neq 0$)
- $f \circ g$ is continuous at a

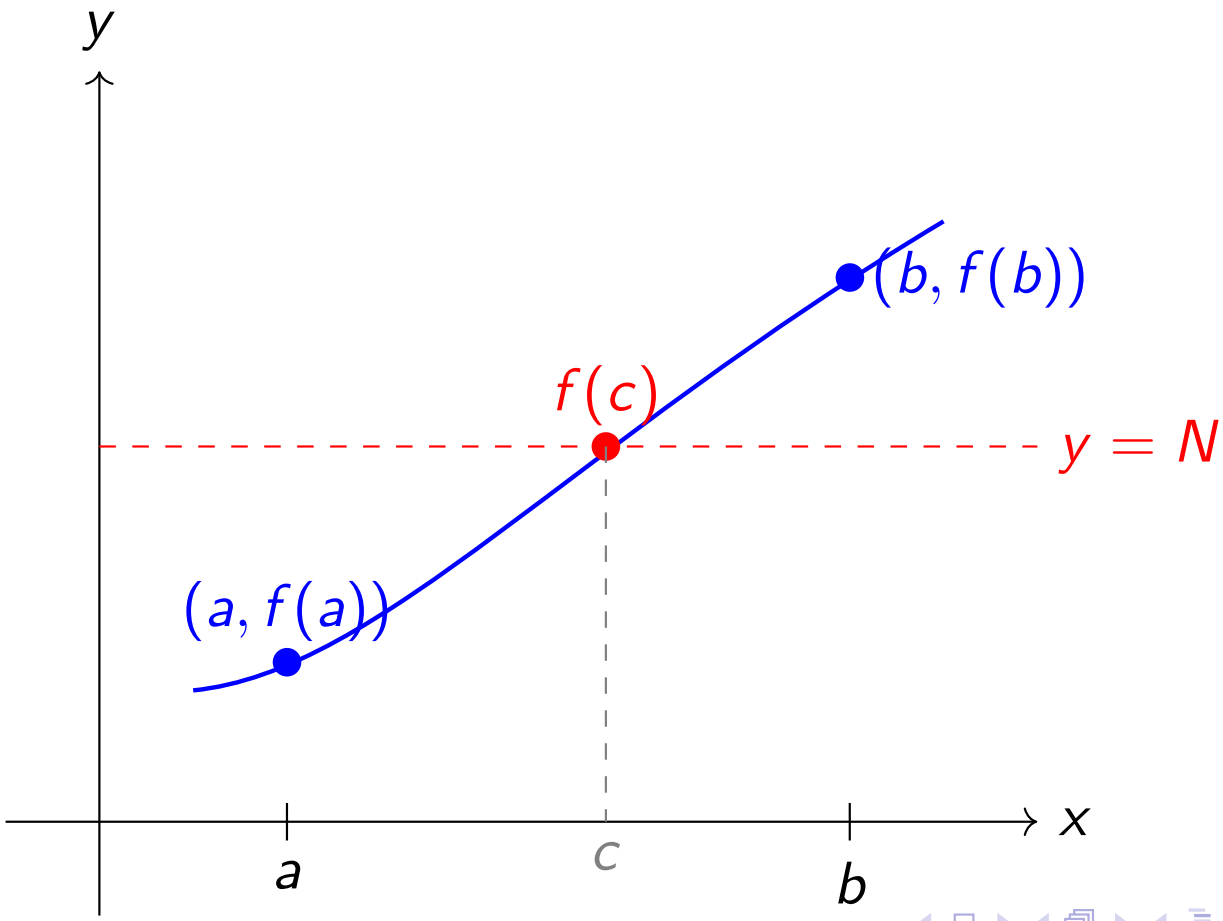
Elementary Continuous Functions

- Polynomials: $p(x) = a_n x^n + \cdots + a_1 x + a_0$
- Rational functions: $\dfrac{p(x)}{q(x)}$ (where $q(x) \neq 0$)
- Trigonometric functions: $\sin x$, $\cos x$, etc.
- Exponential and logarithmic functions
- Root functions

Intermediate Value Theorem (IVT)

Theorem Statement

If f is continuous on $[a, b]$ and N is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in (a, b)$ such that $f(c) = N$.



Practice Problems: Squeeze Theorem

Basic Applications

1. Use the Squeeze Theorem to evaluate: $\lim_{x\to 0} x^3 \sin\left(\frac{1}{x^2}\right)$
2. Evaluate: $\lim_{x\to 0} \frac{1-\cos x}{x^2}$ (Hint: Use $1-\cos x = 2\sin^2(x/2)$)
3. Find: $\lim_{x\to\infty} \frac{\cos x}{x}$

Challenging Problems

1. Prove that $\lim_{x\to 0} x^2\left(1+\sin^2\left(\frac{1}{x}\right)\right) = 0$
2. Evaluate: $\lim_{x\to 0} \sqrt{x^3+x^2}\sin\left(\frac{\pi}{x}\right)$

Practice Problems: Continuity

Continuity Analysis

1. Determine if $f(x) = \frac{x^2-4}{x-2}$ is continuous at $x = 2$
2. Find the value of k that makes $f(x) = \begin{cases} x^2 + 1 & \text{if } x \leq 1 \\ kx - 2 & \text{if } x > 1 \end{cases}$ continuous everywhere
3. Identify all discontinuities of $f(x) = \frac{x}{|x|}$

Intermediate Value Theorem

1. Show that $f(x) = x^3 - 3x + 1$ has a root in the interval $(1, 2)$
2. Prove that the equation $\cos x = x$ has at least one solution

Advanced Practice Problems

Combined Concepts

1. Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$
   - Show that f is continuous at $x = 0$
   - Use Squeeze Theorem to prove continuity
2. Consider $g(x) = \frac{\sin(x-1)}{x-1}$
   - What type of discontinuity occurs at $x = 1$?
   - Can it be removed? If so, how?
   - Use Squeeze Theorem to find the limit as $x \to 1$
3. Prove that if f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then f has at least one root in (a, b)

Real-World Applications

Physics Applications

1. **Position Function:** A car's position $s(t) = t^3 - 6t^2 + 9t$ meters.
   - Is the position function continuous?
   - Use IVT to show the car must be at $s = 4$ meters sometime between $t = 1$ and $t = 3$ seconds

2. **Temperature:** The temperature $T(x)$ along a metal rod is continuous.
   - If $T(0) = 20^\circ C$ and $T(10) = 80^\circ C$, prove the temperature must be $50^\circ C$ somewhere along the rod

Engineering Applications

1. **Signal Processing:** A signal $V(t) = A\sin(\omega t)$ is passed through a limiter that outputs $g(V) = \begin{cases} V & \text{if } |V| \leq 1 \\ \frac{V}{|V|} & \text{if } |V| > 1 \end{cases}$
   - Is $g(V)$ continuous? Identify any discontinuities

Key Concepts Summary

Squeeze Theorem

- Powerful tool for limits involving oscillation or bounds
- Requires finding upper and lower bounding functions
- Both bounds must have the same limit
- Useful for trigonometric limits and oscillatory behavior

Continuity

- Three conditions: $f(a)$ defined, limit exists, limit equals $f(a)$
- Types of discontinuity: removable, jump, infinite, essential
- Algebraic combinations preserve continuity
- Intermediate Value Theorem: Existence of solutions

Problem-Solving Strategies

- For Squeeze Theorem: Find natural bounds ($-1 \leq \sin x \leq 1$, etc.)
- For continuity: Check three conditions systematically
- For IVT: Verify continuity and check function values at endpoints
- Graphical analysis helps understand behavior